

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

NAME: K. HEMA SAI SIDDARTHA

DATE:29/09/2026

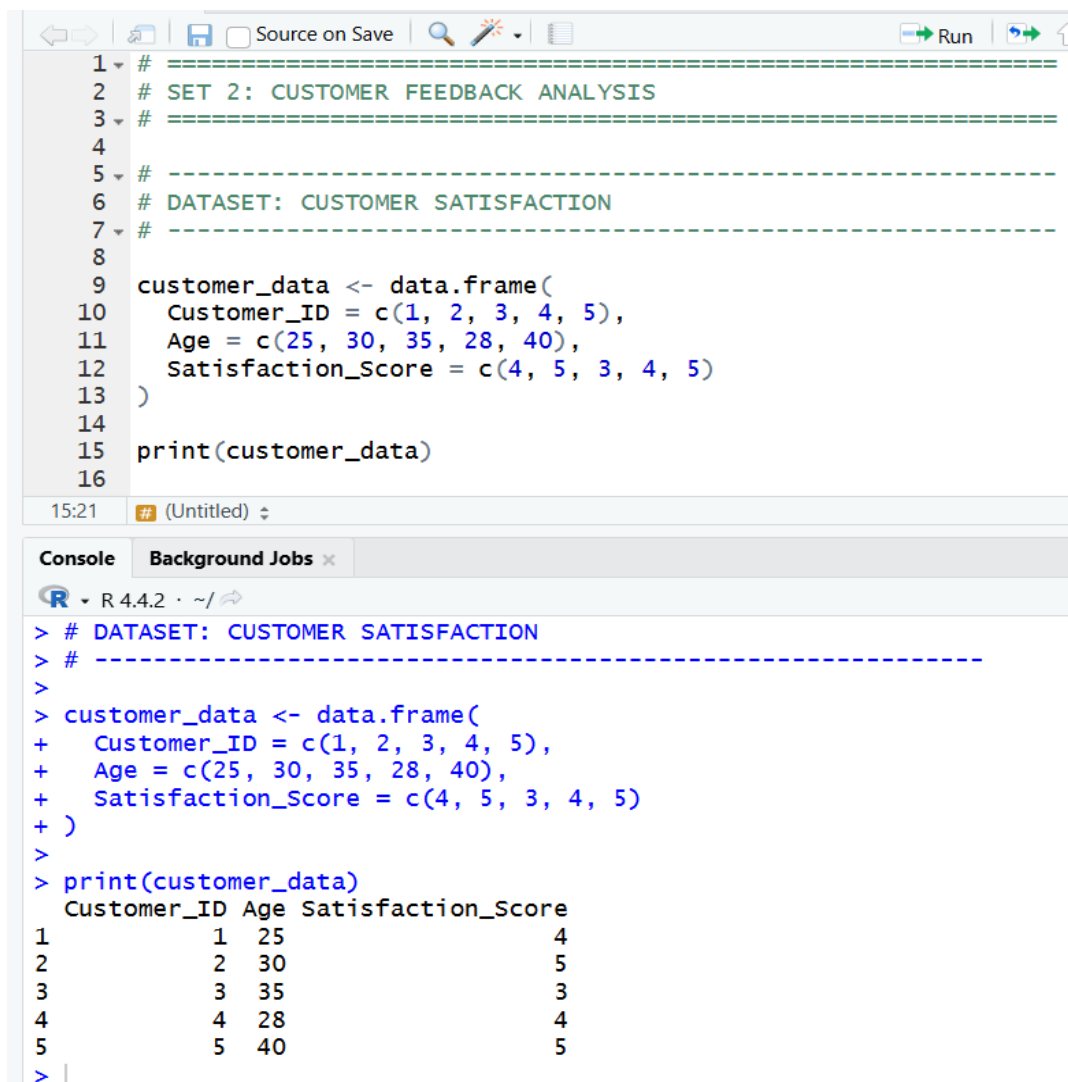
LAB EXP: 2

Reg Mo: 192325036

Course Code/ Name: DSA0601 / Data Handling and Visualization

Set 2: Customer Feedback Analysis

Dataset: Customer Satisfaction



```
1 # =====
2 # SET 2: CUSTOMER FEEDBACK ANALYSIS
3 # =====
4
5 # -----
6 # DATASET: CUSTOMER SATISFACTION
7 # -----
8
9 customer_data <- data.frame(
10   Customer_ID = c(1, 2, 3, 4, 5),
11   Age = c(25, 30, 35, 28, 40),
12   Satisfaction_Score = c(4, 5, 3, 4, 5)
13 )
14
15 print(customer_data)
16
```

15:21 # (Untitled) ↕

Console Background Jobs x

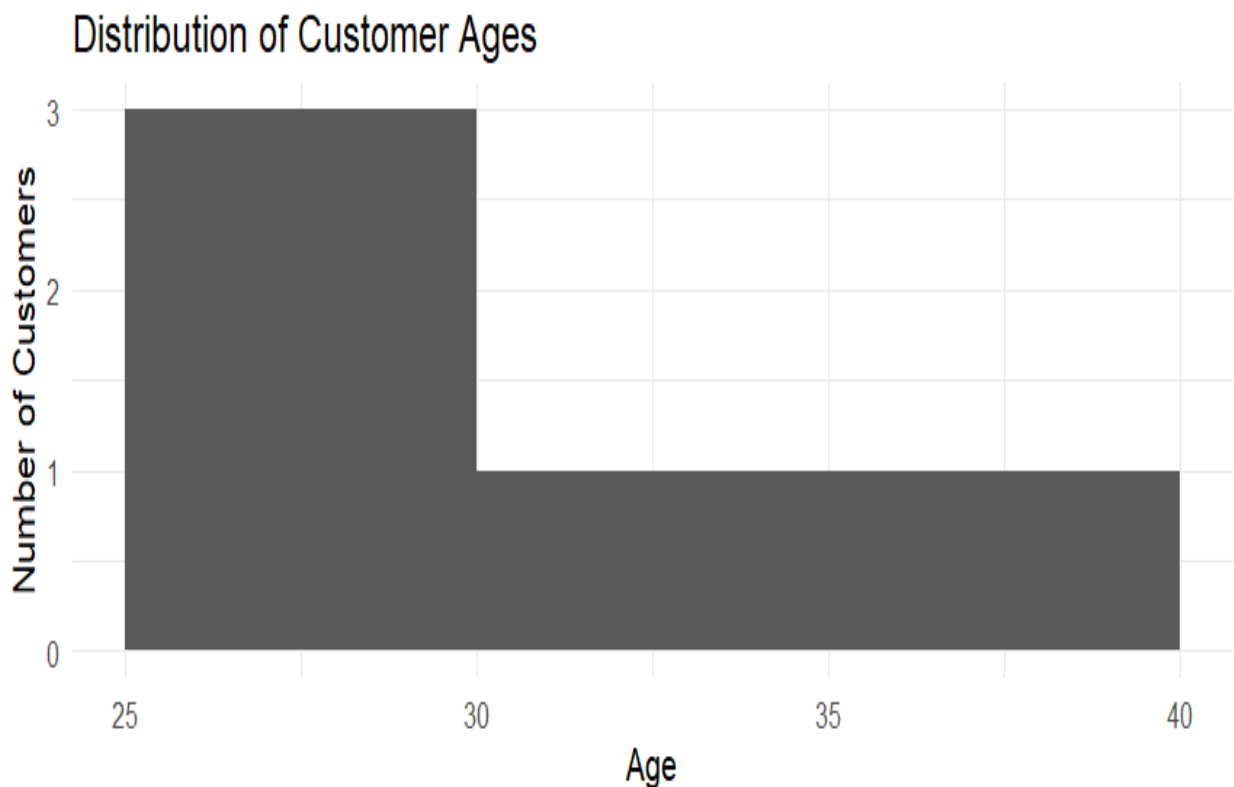
R 4.4.2 · ~/

```
> # DATASET: CUSTOMER SATISFACTION
> # -----
>
> customer_data <- data.frame(
+   Customer_ID = c(1, 2, 3, 4, 5),
+   Age = c(25, 30, 35, 28, 40),
+   Satisfaction_Score = c(4, 5, 3, 4, 5)
+ )
>
> print(customer_data)
  Customer_ID Age Satisfaction_Score
1           1  25                  4
2           2  30                  5
3           3  35                  3
4           4  28                  4
5           5  40                  5
> |
```

1. Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

```
# =====  
# SET 2 - QUESTION 1  
# HISTOGRAM OF CUSTOMER AGES  
# =====  
  
age_histogram <- ggplot(  
  customer_data,  
  aes(x = Age)  
) +  
  geom_histogram(  
    binwidth = 5,  
    boundary = 20  
  ) +  
  labs(  
    title = "Distribution of Customer Ages",  
    x = "Age",  
    y = "Number of Customers"  
  ) +  
  theme_minimal()  
  
print(age_histogram)|
```

OUTPUT:

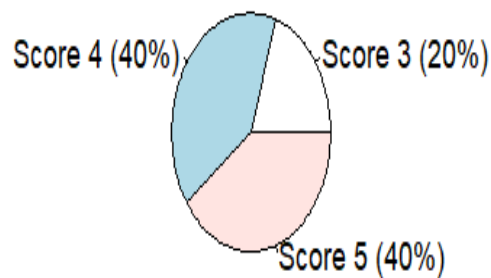


2. Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

```
# =====  
# SET 2 - QUESTION 2  
# PIE CHART  
# Overall Distribution of Customer Satisfaction Scores  
# =====  
# Calculate frequency of satisfaction scores  
satisfaction_data <- customer_data %>%  
  count(Satisfaction_Score)|  
print(satisfaction_data)  
  
# Calculate percentage  
satisfaction_data$Percentage <-  
  satisfaction_data$n /  
  sum(satisfaction_data$n) * 100  
  
# Create labels  
satisfaction_data$Label <- paste0(  
  "Score ",  
  satisfaction_data$Satisfaction_Score,  
  " (" ,  
  round(satisfaction_data$Percentage, 1),  
  "%)"  
)  
  
# Pie chart  
pie(  
  satisfaction_data$n,  
  labels = satisfaction_data$Label,  
  main = "Customer Satisfaction Score Distribution"  
)
```

OUTPUT:

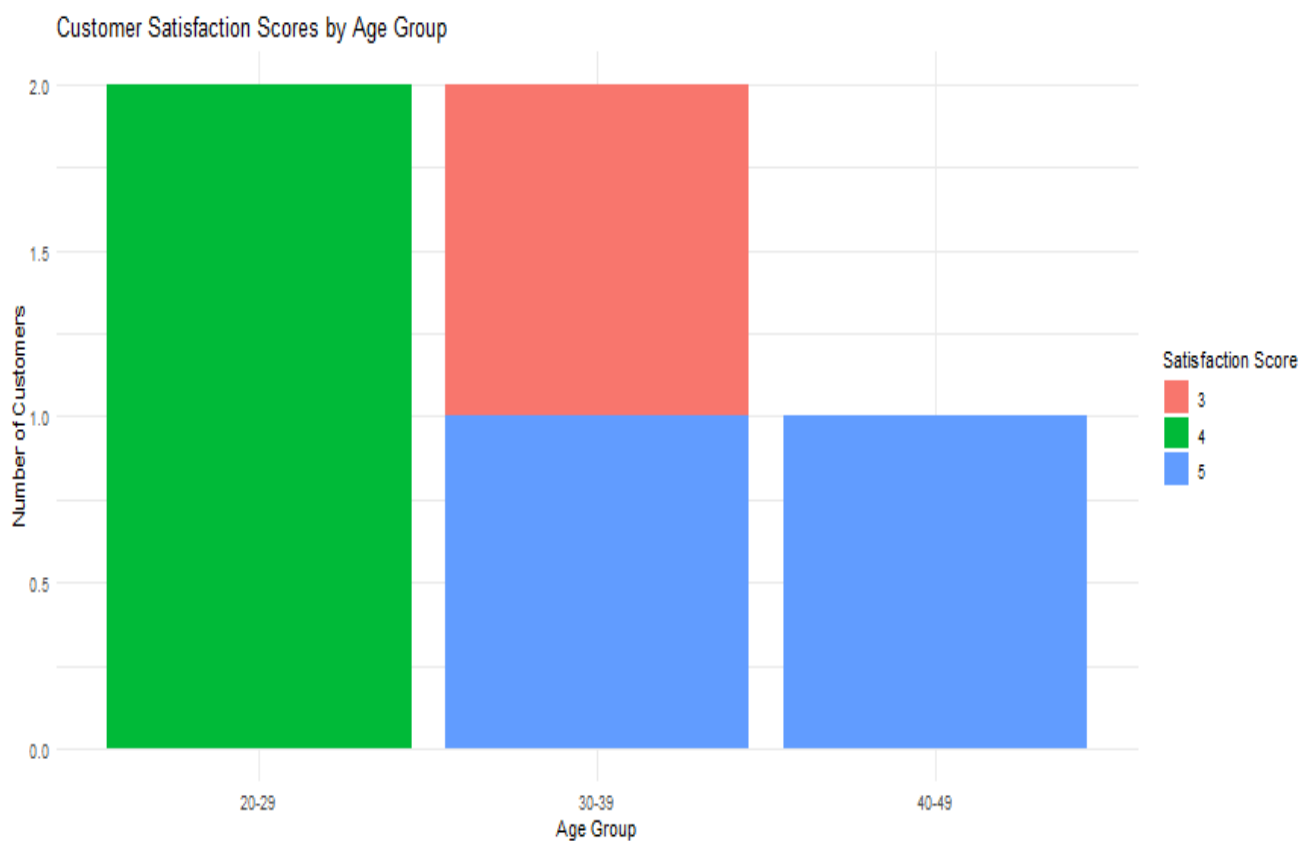
Customer Satisfaction Score Distribution



3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

```
# =====  
# SET 2 - QUESTION 3  
# STACKED BAR CHART  
# Satisfaction Scores by Age Group  
# =====  
  
# Create age groups  
customer_data$Age_Group <- cut(  
  customer_data$Age,  
  breaks = c(20, 29, 39, 49),  
  labels = c(  
    "20-29",  
    "30-39",  
    "40-49"  
  ),  
  include.lowest = TRUE  
)  
  
print(customer_data)  
  
# Convert satisfaction score into factor  
customer_data$Satisfaction_Score <- factor(  
  customer_data$Satisfaction_Score,  
  levels = c(1, 2, 3, 4, 5)  
)  
  
# Stacked bar chart  
stacked_bar <- ggplot(  
  customer_data,  
  aes(  
    x = Age_Group,  
    fill = Satisfaction_Score  
  )  
) +  
  geom_bar() +  
  labs(  
    title = "Customer Satisfaction Scores by Age Group",  
    x = "Age Group",  
    y = "Number of Customers",  
    fill = "Satisfaction Score"  
  ) +  
  theme_minimal()  
print(stacked_bar)
```

OUTPUT:



4. Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

```
# =====  
# SET 2 - QUESTION 4  
# WORD CLOUD  
# Open-Ended Customer Feedback  
# =====  
  
# The original dataset does not contain text feedback.  
# Therefore, sample feedback is created.  
  
feedback <- c(  
  "excellent service and very helpful staff",  
  "good product quality and excellent service",  
  "friendly staff and quick delivery",  
  "product quality is good",  
  "very satisfied with the service",  
  "delivery was fast and service was excellent",  
  "good customer support",  
  "friendly customer support and good quality",  
  "excellent product and fast delivery",  
  "satisfied with product quality"  
)  
  
# Create corpus  
corpus <- Corpus(  
  VectorSource(feedback)  
)  
  
# Convert text to lower case  
corpus <- tm_map(  
  corpus,  
  content_transformer(tolower)  
)  
  
# Remove punctuation  
corpus <- tm_map(  
  corpus,  
  removePunctuation  
)  
  
# Remove numbers  
corpus <- tm_map(  
  corpus,  
  removeNumbers  
)
```

OUTPUT:

